

Algebra II Review: Complex Numbers as Plane Vectors

Context

A complex number behaves like a two-dimensional real vector, but it also has a specially defined multiplication. That multiplication turns each pair $\langle a, b \rangle$ into a single number that can be added, multiplied, divided, measured, and assigned a direction. This viewpoint explains the usual notation $a + bi$ rather than treating it as an arbitrary rule.

Real Pairs with Special Operations

Begin with ordered pairs of real numbers:

$$z = \langle a, b \rangle, \quad w = \langle c, d \rangle.$$

Addition is ordinary vector addition:

$$\langle a, b \rangle + \langle c, d \rangle = \langle a + c, b + d \rangle.$$

Define multiplication by the special rule

$$\boxed{\langle a, b \rangle \langle c, d \rangle = \langle ac - bd, ad + bc \rangle.}$$

This is *not* component-by-component multiplication.

If r is a real scalar, identify it with the pair $\langle r, 0 \rangle$. Multiplication by this pair gives ordinary scalar multiplication:

$$\begin{aligned} \langle r, 0 \rangle \langle a, b \rangle &= \langle ra - 0b, rb + 0a \rangle \\ &= \langle ra, rb \rangle = r \langle a, b \rangle. \end{aligned}$$

Thus real scalar multiplication is a special case of complex multiplication.

Length and Direction

The length, or **modulus**, of $z = \langle a, b \rangle$ is its distance from the origin:

$$|z| = \sqrt{a^2 + b^2}.$$

The **argument** of a nonzero complex number is a direction angle θ measured from the positive real axis:

$$\arg z = \theta, \quad \cos \theta = \frac{a}{|z|}, \quad \sin \theta = \frac{b}{|z|}.$$

The quadrant must agree with the signs of a and b . Arguments differing by $2\pi k$ describe the same direction. In this sheet, $\arg z$ denotes the principal argument in $0 \leq \theta < 2\pi$.

If $r = |z|$, then

$$z = \langle r \cos \theta, r \sin \theta \rangle.$$

From $\langle a, b \rangle$ to $a + bi$

Define

$$1 := \langle 1, 0 \rangle, \quad i := \langle 0, 1 \rangle.$$

Every pair can now be written as

$$\begin{aligned}\langle a, b \rangle &= a\langle 1, 0 \rangle + b\langle 0, 1 \rangle \\ &= a \cdot 1 + b \cdot i \\ &= a + bi.\end{aligned}$$

Therefore

$$\boxed{\langle a, b \rangle \longleftrightarrow a + bi.}$$

The pair and $a + bi$ are two representations of the same object. Geometrically, $a + bi$ is plotted at the point (a, b) in the complex plane. Algebraically, it is one scalar complex number. Complex scalars are therefore “like” two-dimensional real vectors, but their special multiplication gives them additional algebraic structure.

The multiplication rule explains why i is a square root of -1 :

$$\begin{aligned}i^2 &= \langle 0, 1 \rangle \langle 0, 1 \rangle \\ &= \langle 0 \cdot 0 - 1 \cdot 1, 0 \cdot 1 + 1 \cdot 0 \rangle \\ &= \langle -1, 0 \rangle = -1.\end{aligned}$$

Thus $i^2 = -1$, so i is a number whose square is -1 . In the usual Algebra II notation, $i = \sqrt{-1}$.

Using $i^2 = -1$, pair multiplication becomes the familiar calculation

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

Real and Imaginary Parts

For $z = a + bi$,

$$\operatorname{Re}(z) = a, \quad \operatorname{Im}(z) = b.$$

The imaginary part is the real coefficient b , not the term bi . The pair notation records the same information:

$$z = \langle \operatorname{Re}(z), \operatorname{Im}(z) \rangle.$$

Complex Conjugates and Modulus

The **complex conjugate** reflects a complex number across the real axis:

$$z = a + bi \longrightarrow \bar{z} = a - bi.$$

In pair form,

$$\langle a, b \rangle \longrightarrow \langle a, -b \rangle.$$

Multiplying a complex number by its conjugate produces a nonnegative real number:

$$\begin{aligned} z\bar{z} &= (a + bi)(a - bi) \\ &= a^2 - abi + abi - b^2i^2 \\ &= a^2 + b^2 \\ &= |z|^2. \end{aligned}$$

Consequently,

$$\boxed{z\bar{z} = |z|^2}, \quad |z| = |\bar{z}|, \quad z + \bar{z} = 2\operatorname{Re}(z), \quad z - \bar{z} = 2i\operatorname{Im}(z).$$

Example: if $z = 3 - 4i$, then

$$\bar{z} = 3 + 4i, \quad |z| = 5, \quad z\bar{z} = 25.$$

Complex Zeros of Polynomials

Conjugate Pair Theorem

If a polynomial has **real coefficients** and $a + bi$ is a nonreal zero, then its conjugate $a - bi$ is also a zero:

$$P(a + bi) = 0 \implies P(a - bi) = 0.$$

The real-coefficient condition matters. Nonreal zeros of real polynomials occur in conjugate pairs.

The pair produces a real quadratic factor:

$$\begin{aligned} (x - (a + bi))(x - (a - bi)) &= (x - a)^2 + b^2 \\ &= x^2 - 2ax + (a^2 + b^2). \end{aligned}$$

For example, if $2 + 3i$ is a zero of a real polynomial, then $2 - 3i$ is also a zero, and

$$(x - (2 + 3i))(x - (2 - 3i)) = (x - 2)^2 + 9 = x^2 - 4x + 13$$

is a factor.

Fundamental Theorem of Algebra

Every nonconstant polynomial of degree n has exactly n complex zeros when multiplicity is counted. Equivalently, over $\mathbb{C}$,

$$P(x) = a_n(x - r_1)(x - r_2) \cdots (x - r_n).$$

A repeated zero is counted more than once. For example, if

$$P(x) = (x - 3)^2(x^2 + 4),$$

then its four zeros, counting multiplicity, are $3, 3, 2i, -2i$.

Useful consequences:

- A quadratic has two complex zeros, a cubic has three, and a quartic has four, counting multiplicity.
- A real polynomial of odd degree must have at least one real zero because its nonreal zeros pair up.
- For a real polynomial, factor over the reals using linear factors and irreducible quadratic factors; factor over the complex numbers using only linear factors.
- The quadratic formula still applies when $b^2 - 4ac < 0$, using $\sqrt{-m} = i\sqrt{m}$ for $m > 0$.

Reciprocals and Division

For $z = a + bi \neq 0$, multiply by the conjugate to write the reciprocal in $a + bi$ form:

$$\begin{aligned}\frac{1}{z} &= \frac{1}{a + bi} \cdot \frac{a - bi}{a - bi} \\ &= \frac{a - bi}{a^2 + b^2} \\ &= \boxed{\frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2}i}.\end{aligned}$$

Equivalently,

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}.$$

For $w = c + di$ and nonzero $z = a + bi$,

$$\begin{aligned}\frac{w}{z} &= \frac{w\bar{z}}{z\bar{z}} = \frac{(c + di)(a - bi)}{a^2 + b^2} \\ &= \boxed{\frac{ac + bd}{a^2 + b^2} + \frac{ad - bc}{a^2 + b^2}i}.\end{aligned}$$

For example,

$$\frac{2 + i}{1 - 2i} = \frac{(2 + i)(1 + 2i)}{(1 - 2i)(1 + 2i)} = \frac{5i}{5} = i.$$

30-Minute Practice

Instructions. Complete all 12 questions in 30 minutes. Show clear work. Give exact values, write final complex-number answers in $a + bi$ form unless another form is requested, and use $0 \leq \text{Arg } z < 2\pi$.

1. Let $z = \langle 2, -3 \rangle$ and $w = \langle -1, 4 \rangle$.

- (a) Find $2z + w$ in pair form and in $a + bi$ form.
- (b) Find zw using the special pair multiplication, then verify using $a + bi$ notation.

2. Use the pair multiplication rule to show both of the following directly:

$$\langle 0, 1 \rangle^2 = \langle -1, 0 \rangle, \quad \langle r, 0 \rangle \langle a, b \rangle = \langle ra, rb \rangle.$$

Explain what each identity means in $a + bi$ notation.

3. For $z = -\sqrt{3} + i$, find $|z|$ and $\text{Arg } z$. Then write z in polar form

$$|z|(\cos \theta + i \sin \theta).$$

4. Let $z = 2 - 3i$ and $w = -1 + 4i$.

- (a) Find $\bar{z}$, $|z|$, and $z\bar{z}$.
- (b) Write $1/z$ in $a + bi$ form.
- (c) Write w/z in $a + bi$ form.

5. Find the real and imaginary parts of

$$\frac{(3 - 2i)^2}{1 + i}.$$

6. A complex number z satisfies

$$z + \bar{z} = 6, \quad z\bar{z} = 25.$$

Find all possible values of z and the principal argument of each.

7. Solve $x^2 - 6x + 13 = 0$. Factor the quadratic completely over $\mathbb{C}$.

8. A monic quadratic polynomial with real coefficients has $-2 + 3i$ as a zero. Find the other zero and write the polynomial in standard form.

9. Factor completely over $\mathbb{C}$ and list all zeros:

$$x^3 - 5x^2 + 9x - 5.$$

10. Factor completely over the reals and then over $\mathbb{C}$:

$$x^4 - 8x^3 + 26x^2 - 48x + 45.$$

State every zero and its multiplicity.

11. Find the real quartic polynomial $P(x)$ of least degree such that 2 is a zero of multiplicity 2, $-1 + 2i$ is a zero, and $P(0) = 40$. Write $P(x)$ in factored and standard forms.

12. The polynomial

$$P(x) = x^3 + ax^2 + bx + 10$$

has real coefficients, and $1 + 2i$ is a zero. Find a , b , and all three zeros of P .